

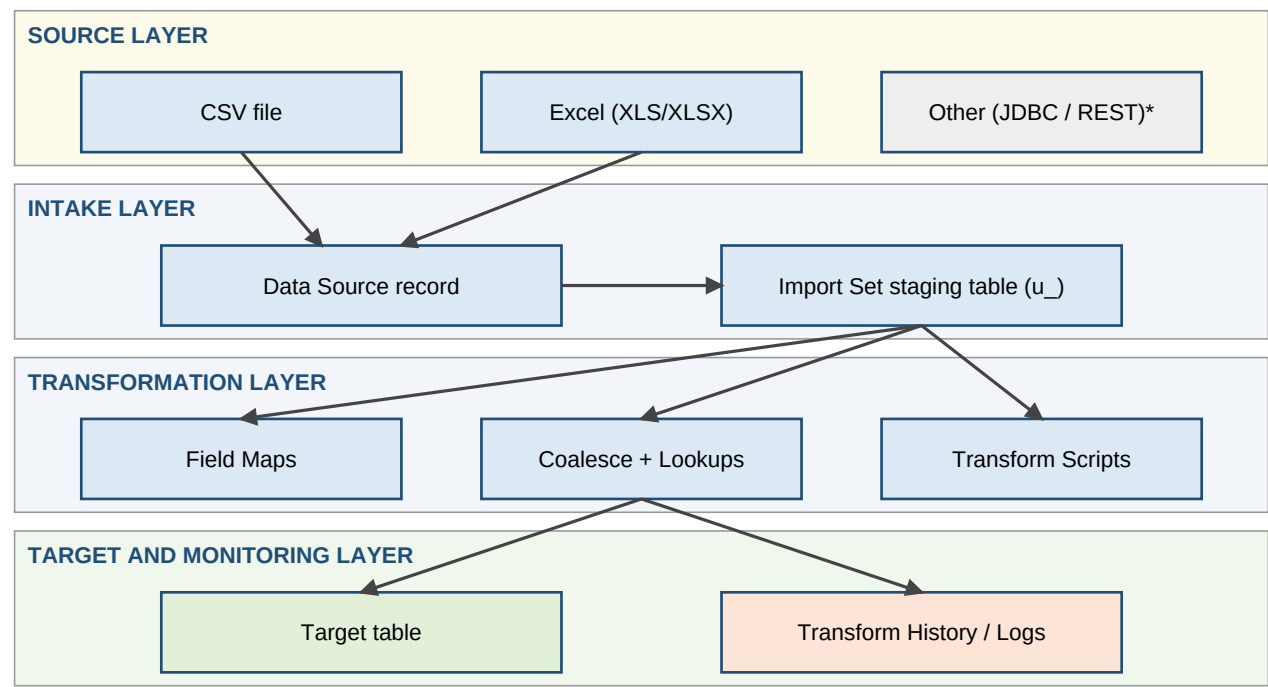
Solution Architecture

Project Design Phase | Import Data Using Transform Maps

1. Architecture Overview

The solution has four layers. Data enters from files, is staged in an import set table, is processed by a Transform Map, and is stored in the target table while results are written to logs.

2. Architecture Diagram



** Planned for a later phase (web service / JDBC imports).*

3. Component Description

Component	Description	Responsibility
Data Source	Record with file type, sheet number, header row	Defines how the file is read
Import Set table	Staging table created from file headers	Holds raw rows safely
Transform Map	Rules linking staging to target	Controls mapping and processing
Field Maps	One source to one target field links	Move values into fields
Coalesce	Unique key match	Update existing or insert new
Transform Scripts	onStart, onBefore, onAfter, field scripts	Clean, validate, follow-up actions
Target table	Live ServiceNow table	Stores final data
Transform History / Logs	System log tables	Record counts and errors

4. Design Decisions

Decision	Reason
Use a staging table	Raw data never touches live tables
One Transform Map per source layout	Keeps mapping clear and maintainable
Coalesce on a stable unique key	Prevents duplicates on re-load
Scripts only where needed	Direct mapping is simpler and faster
Test in sub-production first	Reduces risk to live data

5. Deployment View

- Development in a Personal Developer Instance (PDI).
- Configuration captured in an update set for moving to other instances.
- Source files are uploaded by an admin with the import_transformer role.